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Operator Jensen's Inequality on C^*-algebras

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Abstract A real continuous function which is defined on an interval is said to be $\mathfrak{A}$-convex if it is convex on the set of self-adjoint elements, with spectra in the interval, in all matrix algebras of the unital C^*-algebra $\mathfrak{A}$. We give a general formation of Jensen's inequality for $\mathfrak{A}$-convex functions.

Keywords Jensen's inequality, C^*-algebra, $\mathfrak{A}$-convex function, Bochner integral, unital $\mathfrak{A}$-field

MR(2010) Subject Classification 46L05, 46L07, 47A63

1 Introduction

Classic Jensen's inequality says that if Φ is a convex function from (a,b) to $\mathbb{R}$, then

$$\Phi\bigg(\sum_{i=1}^{m}\lambda_i x_i\bigg)\le\sum_{i=1}^{m}\lambda_i\Phi(x_i),$$

whenever $\lambda_i\ge 0$ for $i=1,2,\dots,m$ with $\sum_{i=1}^{m}\lambda_i=1$ and $x_i\in(a,b)$ for all i (see [11, 12]). Let (X,Σ,μ) be a measure space with $\mu(X)=1$. If $f:X\mapsto\mathbb{R}$ is an integrable function with $f(X)\subseteq(\alpha,\beta)$, and Φ is a convex function from (α,β) to $\mathbb{R}$, then

$$\Phi\bigg(\int_X f d\mu\bigg)\le\int_X(\Phi\circ f)d\mu.$$

This is the continuous version of Jensen's inequality.

Various extensions of Jensen's inequality to more general context have been considered for a number of decades (see [1, 3, 4, 6, 8–10, 13, 14]). If f is a real continuous function on some interval I in $\mathbb{R}$ and x is a bounded self-adjoint operator on a Hilbert space with spectrum in I, we can use spectral theory to define a bounded self-adjoint operator

$$f(x)=\int_{-\infty}^{\infty}f(\lambda)dE_x(\lambda),$$

where E_x is the spectral measure of x. A continuous function $\Phi:I\mapsto\mathbb{R}$ is said to be *operator convex* if

$$\Phi(\alpha x+(1-\alpha)y)\le\alpha\Phi(x)+(1-\alpha)\Phi(y)$$

for any $\alpha\in[0,1]$ and every pair of self-adjoint operators x and y on an infinite-dimensional Hilbert space H with spectra in I. In [10], Hansen and Pedersen proved that if Φ is a continuous,

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operator convex function defined on an interval I, then for each natural number n, Jensen's operator inequality

$$\Phi\left(\sum_{i=1}^{n} a_i^* x_i a_i\right) \leq \sum_{i=1}^{n} a_i^* \Phi(x_i) a_i$$

holds for every n-tuple $(x_1, \ldots, x_n)$ of bounded, self-adjoint operators on an arbitrary Hilbert space H with spectra contained in I and every n-tuple $(a_1, \ldots, a_n)$ of operators on H with $\sum_{i=1}^{n} a_i^* a_i = 1_{B(H)}$. In [8], a Bochner integral formation of Jensen's inequality is presented for self-adjoint matrix-valued functions and measures. Farenick and Zhou proved that for a matrix-valued probability measure space (X, Σ, v), if I is an open interval such that $[\alpha, \beta] \subset I$ and if f is a measurable map from X to the real vector space of all self-adjoint operators on $\mathbb{C}^n$ for which the spectra $\sigma(f(x)) \subseteq [\alpha, \beta]$ for every $x \in X$, then

$$\Phi\left(\int_X f dv\right) \leq \int_X (\Phi \circ f)\, dv$$

for every operator convex function $\Phi : I \mapsto \mathbb{R}$.

The aim of this paper is to generalize Jensen's inequality to (abstract) unital C^*-algebras. In Section 2, we define the $\mathfrak{A}$-convexity of a real continuous function, and a discrete Jensen's inequality is given. Using the Bochner integral on vector-valued functions, we obtain a continuous version of Jensen's inequality in Section 3. Based on the results in Sections 2 and 3, we give a reformation of Jensen's inequality, in Section 4, in the $\mathfrak{A}$-convexity which contains results above as special cases.

2 Discrete Operator Jensen's Inequality on C^*-algebras

Definition 2.1 *Let $\mathfrak{A}$ be a C^*-algebra, and let I be an interval in $\mathbb{R}$. If Φ is a continuous function from I to $\mathbb{R}$, and a is a self-adjoint element in $\mathfrak{A}$ with spectrum $\sigma(a) \subseteq I$, then $\Phi(a) \in \mathfrak{A}$ is defined via functional calculus. A continuous function $\Phi : I \mapsto \mathbb{R}$ is said to be $\mathfrak{A}$-convex of order n if for any self-adjoint $a, b \in M_n(\mathfrak{A})$ with $\sigma(a), \sigma(b) \subseteq I$, the inequality*

$$\Phi(\lambda a + (1-\lambda)b) \leq \lambda \Phi(a) + (1-\lambda)\Phi(b)$$

holds for all $\lambda \in [0, 1]$. If for any $n \in \mathbb{N}$ the function Φ is $\mathfrak{A}$-convex of order n, we say that Φ is $\mathfrak{A}$-convex.

Theorem 2.2 *Let $\mathfrak{A}$ be a unital C^*-algebra with unit $1_{\mathfrak{A}}$, and let I be an interval in $\mathbb{R}$. If $\Phi : I \mapsto \mathbb{R}$ is an $\mathfrak{A}$-convex function and if self-adjoint elements $a_1, \ldots, a_n$ in $\mathfrak{A}$ have spectra contained in I, then*

$$\Phi\left(\sum_{i=1}^{n} b_i^* a_i b_i\right) \leq \sum_{i=1}^{n} b_i^* \Phi(a_i) b_i$$

for any $b_1, \ldots, b_n \in \mathfrak{A}$ that satisfy $\sum_{i=1}^{n} b_i^* b_i = 1_{\mathfrak{A}}$.

Proof Choose $\alpha, \beta \in \mathbb{R}$ such that $\sigma(a_i) \subseteq [\alpha, \beta] \subseteq I$ for $i = 1, 2, \ldots, n$. Since $a_1, \ldots, a_n$ are self-adjoint, we have

$$\alpha 1_{\mathfrak{A}} \leq a_i \leq \beta 1_{\mathfrak{A}}, \quad i = 1, 2, \ldots, n.$$

So we have

$$\alpha 1_{\mathfrak{A}} \leq \sum_{i=1}^{n} b_i^* a_i b_i \leq \beta 1_{\mathfrak{A}}.$$

Hence $\sum_{i=1}^{n} b_i^* a_i b_i$ is a self-adjoint element with spectrum $\sigma(\sum_{i=1}^{n} b_i^* a_i b_i) \subseteq [\alpha, \beta] \subseteq I$. Denote

$$u = \begin{pmatrix} 1_{\mathfrak{A}} - b_1 b_1^* & -b_1 b_2^* & \cdots & -b_1 b_n^* & b_1 \\ -b_2 b_1^* & 1_{\mathfrak{A}} - b_2 b_2^* & \cdots & -b_2 b_n^* & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ -b_n b_1^* & -b_n b_2^* & \cdots & 1_{\mathfrak{A}} - b_n b_n^* & b_n \\ -b_1^* & -b_2^* & \cdots & -b_n^* & 0 \end{pmatrix}$$

and

$$a = \begin{pmatrix} a_1 & 0 & \cdots & 0 & 0 \\ 0 & a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & a_n & 0 \\ 0 & 0 & \cdots & 0 & a_n \end{pmatrix}.$$

Then $u, a \in M_{n+1}(\mathfrak{A})$, u is a unitary element in $M_{n+1}(\mathfrak{A})$, and

$$u^* a u = \begin{pmatrix} * & * \\ * & \sum_{i=1}^{n} b_i^* a_i b_i \end{pmatrix}.$$

Set $\theta = \exp\left(\frac{2\pi \mathrm{i}}{n+1}\right)$, and denote

$$v = \begin{pmatrix} \theta 1_{\mathfrak{A}} & 0 & \cdots & 0 & 0 \\ 0 & \theta^2 1_{\mathfrak{A}} & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & \theta^n 1_{\mathfrak{A}} & 0 \\ 0 & 0 & \cdots & 0 & 1_{\mathfrak{A}} \end{pmatrix}.$$

Then v is also a unitary element in $M_{n+1}(\mathfrak{A})$. By computation, we have

$$\frac{1}{n+1} \sum_{k=1}^{n+1} v^{-k} u^* a u v^k = \begin{pmatrix} * & 0 \\ 0 & \sum_{i=1}^{n} b_i^* a_i b_i \end{pmatrix}.$$

From the properties of functional calculus in C^*-algebras (see [5]), we have

$$\Phi(a) = \begin{pmatrix} \Phi(a_1) & 0 & \cdots & 0 & 0 \\ 0 & \Phi(a_2) & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & \Phi(a_n) & 0 \\ 0 & 0 & \cdots & 0 & \Phi(a_n) \end{pmatrix}.$$

Similarly, we obtain

$$\frac{1}{n+1}\sum_{k=1}^{n+1} v^{-k}u^*\Phi(a)uv^k = \begin{pmatrix} * & 0 \\ 0 & \sum_{i=1}^n b_i^*\Phi(a_i)b_i \end{pmatrix}.$$

Using the $\mathfrak{A}$-convexity of Φ and the properties of functional calculus in C^*-algebras, we have

$$\begin{aligned}\begin{pmatrix} \Phi(*) & 0 \\ 0 & \Phi(\sum_{i=1}^n b_i^*a_ib_i) \end{pmatrix} &= \Phi\begin{pmatrix} * & 0 \\ 0 & \sum_{i=1}^n b_i^*a_ib_i \end{pmatrix} = \Phi\Big(\frac{1}{n+1}\sum_{k=1}^{n+1} v^{-k}u^*auv^k\Big) \\ &\le \frac{1}{n+1}\sum_{k=1}^{n+1}\Phi(v^{-k}u^*auv^k) = \frac{1}{n+1}\sum_{k=1}^{n+1} v^{-k}u^*\Phi(a)uv^k \\ &= \begin{pmatrix} * & 0 \\ 0 & \sum_{i=1}^n b_i^*\Phi(a_i)b_i \end{pmatrix}.\end{aligned}$$

Therefore, $\Phi\left(\sum_{i=1}^n b_i^*a_ib_i\right) \le \sum_{i=1}^n b_i^*\Phi(a_i)b_i$. □

The following example indicates that in Theorem 2.2 above, we cannot replace the $\mathfrak{A}$-convexity of Φ by the convexity of Φ only on $\mathfrak{A}$.

Example 2.3 Let $\mathfrak{A} = M_2$ and set

$$\Phi(t) = t + \frac{1}{2}t^2 + \frac{1}{3}t^3 - \log(1+t), \quad t \in [0, 0.1).$$

Then from the proof of Theorem 5.8 in [15], we see that, for every $\lambda \in [0,1]$ and every pair of self-adjoint elements $a, b \in \mathfrak{A}$ with spectra contained in $[0, 0.1)$, the inequality

$$\Phi(\lambda a + (1-\lambda)b) \le \lambda\Phi(a) + (1-\lambda)\Phi(b)$$

holds. But $g(t) = \frac{\Phi(t)}{t}$, $t \in (0, 0.1)$, is not monotone on $\mathfrak{A}$ (see [15]). By [16, Theorem 2.1], there exist a positive element $a_0 \in \mathfrak{A}$ with its spectrum in $[0, 0.1)$ and an element $c_0 \in \mathfrak{A}$ satisfying $c_0^*c_0 \le 1_{\mathfrak{A}}$ such that

$$\Phi(c_0^*a_0c_0) \nleq c_0^*\Phi(a_0)c_0.$$

Denote $c_1 = \sqrt{1_{\mathfrak{A}} - c_0^*c_0}$. Then $c_1 \in \mathfrak{A}$ and $c_0^*c_0 + c_1^*c_1 = 1_{\mathfrak{A}}$. Therefore, we have

$$\Phi(c_0^*a_0c_0 + c_1^*0c_1) = \Phi(c_0^*a_0c_0) \nleq c_0^*\Phi(a_0)c_0 = c_0^*\Phi(a_0)c_0 + c_1^*\Phi(0)c_1.$$

Corollary 2.4 *Let $\mathfrak{A}$ be a unital C^*-algebra with unit $1_{\mathfrak{A}}$, and let I be an interval in $\mathbb{R}$ such that $0 \in I$. If $\Phi : I \mapsto \mathbb{R}$ is an $\mathfrak{A}$-convex function and $\Phi(0) \le 0$, then for any self-adjoint elements $a_1, \ldots, a_n$ in $\mathfrak{A}$ with spectra contained in I, we have*

$$\Phi\Big(\sum_{i=1}^n b_i^*a_ib_i\Big) \le \sum_{i=1}^n b_i^*\Phi(a_i)b_i$$

for any $b_1, \ldots, b_n \in \mathfrak{A}$ that satisfy $\sum_{i=1}^n b_i^*b_i \le 1_{\mathfrak{A}}$.

Proof Denote $b_{n+1} = (1_{\mathfrak{A}} - \sum_{i=1}^n b_i^*b_i)^{\frac{1}{2}}$. Then we have that $b_{n+1} \in \mathfrak{A}$ and $\sum_{i=1}^{n+1} b_i^*b_i = 1_{\mathfrak{A}}$. Take $a_{n+1} = 0$. By Theorem 2.2, we obtain

$$\Phi\Big(\sum_{i=1}^n b_i^*a_ib_i\Big) = \Phi\Big(\sum_{i=1}^{n+1} b_i^*a_ib_i\Big) \le \sum_{i=1}^{n+1} b_i^*\Phi(a_i)b_i$$

$$= \sum_{i=1}^{n} b_i^* \Phi(a_i) b_i + b_{n+1}^* \Phi(0) b_{n+1}$$
$$\leq \sum_{i=1}^{n} b_i^* \Phi(a_i) b_i. \qquad \square$$

Corollary 2.5 *Let $\mathfrak{A}$ be a unital C^*-algebra with unit $1_{\mathfrak{A}}$, and let I be an interval in $\mathbb{R}$. If $\Phi : I \mapsto \mathbb{R}$ is an $\mathfrak{A}$-convex function and if self-adjoint elements $a_1, \ldots, a_n$ in $M_k(\mathfrak{A})$ have spectra contained in I, then*

$$\Phi\left(\sum_{i=1}^{n} b_i^* a_i b_i\right) \leq \sum_{i=1}^{n} b_i^* \Phi(a_i) b_i$$

for any $b_1, \ldots, b_n \in M_k(\mathfrak{A})$ that satisfy $\sum_{i=1}^{n} b_i^* b_i = \underbrace{1_{\mathfrak{A}} \oplus \cdots \oplus 1_{\mathfrak{A}}}_{k}$.

Proof This follows from the $M_k(\mathfrak{A})$-convexity of Φ. $\square$

We recall that a continuous function Φ from an interval I to $\mathbb{R}$ is said to be *matrix convex of order n* if

$$\Phi(\alpha x + (1-\alpha)y) \leq \alpha \Phi(x) + (1-\alpha)\Phi(y)$$

for any $\alpha \in [0,1]$ and every pair of self-adjoint operators x and y on a Hilbert space of finite dimension n with spectra contained in I. From [2, Lemma 3.1], we see that a function is operator convex if and only if it is matrix convex of arbitrary order. So an operator convex function must be $B(H)$-convex for any Hilbert space H, and we obtain the following result in [10].

Corollary 2.6 *If Φ is a continuous, operator convex function defined on an interval I, then for each natural number n, Jensen's operator inequality*

$$\Phi\left(\sum_{i=1}^{n} a_i^* x_i a_i\right) \leq \sum_{i=1}^{n} a_i^* \Phi(x_i) a_i$$

holds for every n-tuple $(x_1, \ldots, x_n)$ of bounded, self-adjoint operators on an arbitrary Hilbert space H with spectra contained in I and every n-tuple $(a_1, \ldots, a_n)$ of operators on H with $\sum_{i=1}^{n} a_i^* a_i = 1_{B(H)}$.

3 Continuous Operator Jensen's Inequality on C^*-algebras

If (X, Σ, μ) is a measure space and B is a Banach space, a map $f : X \mapsto B$ is called *simple* if there exist $b_1, \ldots, b_n \in B$ and $E_1, \ldots, E_n \in \Sigma$ which satisfy that $E_i \cap E_j = \emptyset$ for $i \neq j$, such that

$$f(x) = \sum_{i=1}^{n} b_i \chi_{E_i}(x), \quad x \in X,$$

where $\chi_{E_i}(x) = 1$ if $x \in E_i$ and $\chi_{E_i}(x) = 0$ if $x \notin E_i$. A map $f : X \mapsto B$ is called *μ-measurable* if there exists a sequence of simple maps $\{f_n\}$ from X to B with

$$\lim_{n \to \infty} \|f_n(x) - f(x)\| = 0$$

μ-almost everywhere. A map $f : X \mapsto B$ is called *weakly μ-measurable* if for each $\phi \in B^*$ the function $\phi(f)$ is μ-measurable, where B^* is the dual space of B. By Pettis's measurability theorem, a μ-measurable map from a measure space to a Banach space is weakly μ-measurable [7].

Let $\mathfrak{A}$ be a unital C^*-algebra. For $b^* = b \in \mathfrak{A}$, we denote

$$(-\infty, b] = \{a \in \mathfrak{A} : a^* = a, b - a \geq 0\},$$

$$[b, +\infty) = \{a \in \mathfrak{A} : a^* = a, a - b \geq 0\}.$$

Furthermore, if b and c are self-adjoint elements in $\mathfrak{A}$ with $b \geq c$, then let

$$[c, b] = \{a \in \mathfrak{A} : a^* = a, b \geq a, a \geq c\}.$$

A *Lip-norm* on a unital C^*-algebra $\mathfrak{A}$ with unit $1_{\mathfrak{A}}$ is a seminorm L on $\mathfrak{A}$, possibly taking the value $+\infty$, such that $L(1_{\mathfrak{A}}) = 0$; $L(a^*) = L(a)$ for $a \in \mathfrak{A}$; the metric ρ_L on the state space $S(\mathfrak{A})$ of $\mathfrak{A}$ given by $\rho_L(\varphi, \psi) = \sup\{|\varphi(a) - \psi(a)| : L(a) \leq 1\}$ induces the weak*-topology on $S(\mathfrak{A})$ (see [17, 20, 21]).

Proposition 3.1 *Let $\mathfrak{A}$ be a unital C^*-algebra with a Lip-norm L. If (X, Σ, μ) is a measure space and f is a μ-measurable map from X to the real vector space $\mathfrak{A}_{\rm sa}$ of self-adjoint elements in $\mathfrak{A}$, then*

(1) *$f^{-1}([b, +\infty)) \in \Sigma$ for every $b \in \mathfrak{A}_{\rm sa}$;*

(2) *$f^{-1}((-\infty, b]) \in \Sigma$ for every $b \in \mathfrak{A}_{\rm sa}$;*

(3) *$f^{-1}([c, b]) \in \Sigma$ for any elements b and c in $\mathfrak{A}_{\rm sa}$ with $b \geq c$.*

Proof Since $f(x)$ is μ-measurable, by the property of measurable function, we have

$$\{x \in X : \varphi(f(x) - b) \geq 0\}$$

is a measurable set for $b \in \mathfrak{A}_{\rm sa}$ and $\varphi \in S(\mathfrak{A})$. Since L is a Lip-norm on $\mathfrak{A}$, $S(\mathfrak{A})$ is a compact set in the topology induced by the metric ρ_L. So $S(\mathfrak{A})$ is separable. Let $\{\varphi_i\}_{i=1}^{\infty}$ be a dense subset of $S(\mathfrak{A})$. Then $f(x) \geq b$ if and only if $\varphi_i(f(x) - b) \geq 0$ for each φ_i. So

$$f^{-1}([b, +\infty)) = \bigcap_{i=1}^{\infty} \{x \in X : \varphi_i(f(x) - b) \geq 0\}.$$

From the measurability of each set $\{x \in X : \varphi_i(f(x) - b) \geq 0\}$, we see that $f^{-1}([b, +\infty)) \in \Sigma$, and this proves (1).

In a similar way, we have that $f^{-1}((-\infty, b]) \in \Sigma$ for (2) and $f^{-1}([c, b]) \in \Sigma$ for (3). □

Let (X, Σ, μ) be a measure space, and let B be a Banach space. A μ-measurable map $f : X \mapsto B$ is said to be *Bochner integrable* if there exists a sequence of simple maps $\{f_n\}$ from X to B such that

$$\lim_{n \to \infty} \int_X \|f_n(x) - f(x)\| d\mu(x) = 0.$$

In this case, for any $E \in \Sigma$, the *Bochner integral* of f over E is defined by

$$\int_E f(x) d\mu(x) = \lim_{n \to \infty} \int_E f_n(x) d\mu(x),$$

in the sense of strong convergence in B, where $\int_E f_n(x)d\mu(x)$ is defined in an obvious way [7]. By [7, Chapter II, Theorem 2], a μ-measurable function $f : X \mapsto B$ is Bochner integrable if and only if $\int_X \|f\|d\mu < \infty$.

Lemma 3.2 *Let (X, Σ, μ) be a measure space, and let $\mathfrak{A}$ be a C^*-algebra. If the maps $f, g : X \mapsto \mathfrak{A}_{\rm sa}$ are Bochner integrable, then*

(1) $\int_X (f+g)d\mu = \int_X fd\mu + \int_X gd\mu$;

(2) $\varphi\left(\int_X fd\mu\right) = \int_X \varphi(f)d\mu$ *for each* $\varphi \in \mathfrak{A}^*$;

(3) *for any* $a \in \mathfrak{A}$, a^*fa *is Bochner integrable and*

$$a^*\left(\int_X fd\mu\right)a = \int_X a^*fad\mu;$$

(4) $\int_X fd\mu \le \int_X gd\mu$ *if* $f(x) \le g(x)$ *for all* $x \in X$.

Proof (1) is clear by the definition of Bochner integral, and (2) follows from the continuity of φ.

Since f is Bochner integrable, we have $\int_X \|f\|d\mu < \infty$. So

$$\int_X \|a^*fa\|d\mu \le \|a\|^2 \int_X \|f\|d\mu < \infty,$$

and hence $a^*f(x)a$ is Bochner integrable. Let $\{f_n\}$ be a sequence of simple maps from X to $\mathfrak{A}_{\rm sa}$ such that $\lim_{n\to\infty} \int_X \|f_n(x) - f(x)\|d\mu(x) = 0$. Assume that

$$f_n = \sum_{i=1}^{m_n} b_{n,i}\chi_{E_{n,i}}, \quad n = 1, 2, \ldots,$$

where $b_{n,i} \in \mathfrak{A}_{\rm sa}$ and $E_{n,i} \in \Sigma$ with $E_{n,i} \cap E_{n,j} = \emptyset$ for $i \ne j$. We have

$$a^*\left(\int_X f_nd\mu\right)a = a^*\left(\sum_{i=1}^{m_n} b_{n,i}\mu(E_{n,i})\right)a = \sum_{i=1}^{m_n} a^*b_{n,i}a\mu(E_{n,i}) = \int_X a^*f_nad\mu.$$

It is clear that

$$\lim_{n\to\infty} a^*\left(\int_X f_nd\mu\right)a = a^*\left(\int_X fd\mu\right)a \quad \text{and} \quad \lim_{n\to\infty}\int_X a^*f_nad\mu = \int_X a^*fad\mu.$$

So we get (3).

It remains to prove (4). For each $\varphi \in S(\mathfrak{A})$, by (2), we have

$$\varphi\left(\int_X fd\mu\right) = \int_X \varphi(f)d\mu \quad \text{and} \quad \varphi\left(\int_X gd\mu\right) = \int_X \varphi(g)d\mu.$$

Since $f(x) \le g(x)$ for all $x \in X$, we have $\varphi(f(x)) \le \varphi(g(x))$ for all $x \in X$, and hence $\int_X \varphi(f)d\mu \le \int_X \varphi(g)d\mu$. So we obtain

$$\varphi\left(\int_X fd\mu\right) = \int_X \varphi(f)d\mu \le \int_X \varphi(g)d\mu = \varphi\left(\int_X gd\mu\right).$$

Because φ is arbitrary, we get $\int_X fd\mu \le \int_X gd\mu$. □

Lemma 3.3 *Let (X, Σ, μ) be a measure space, and let $\mathfrak{A}$ be a unital C^*-algebra. Suppose that I is an interval in $\mathbb{R}$ which contains $[\alpha, \beta]$, where $\alpha, \beta \in \mathbb{R}$. If $\{a_n\}$ is a sequence of elements in*

$\mathfrak{A}_{\rm sa}$ *with limit* $\lim_{n\to\infty} a_n = a \in \mathfrak{A}$ *and spectra* $\sigma(a_n) \subseteq [\alpha, \beta]$ *for all* $n \in \mathbb{N}$, *then* $\sigma(a) \subseteq [\alpha, \beta]$ *and*

$$\lim_{n\to\infty} \|\Phi(a_n) - \Phi(a)\| = 0$$

for every $\mathfrak{A}$*-convex function* $\Phi : I \mapsto \mathbb{R}$ *of order one.*

Proof It is clear that a is self-adjoint. Since the spectrum is upper semicontinuous and the spectra $\sigma(a_n) \subseteq [\alpha, \beta]$ for all $n \in \mathbb{N}$, we have that $\sigma(a) \subseteq [\alpha, \beta]$.

For any polynomial $p(t) = \sum_{k=0}^{m} \lambda_k t^k$, we have

$$\begin{aligned}\|p(a_n) - p(a)\| &= \left\| \sum_{k=0}^{m} \lambda_k a_n^k - \sum_{k=0}^{m} \lambda_k a^k \right\| = \left\| \sum_{k=1}^{m} \lambda_k (a_n^k - a^k) \right\| \\ &= \left\| \sum_{k=1}^{m} \lambda_k \sum_{i=0}^{k-1} a^i (a_n - a) a_n^{k-i-1} \right\| \\ &\le \|a_n - a\| \sum_{k=1}^{m} \sum_{i=0}^{k-1} |\lambda_k| \|a\|^i \|a_n\|^{k-i-1}.\end{aligned}$$

Since $\{a_n\}$ is convergent in the norm topology, $\{\|a_n\|\}$ is bounded. So $\lim_{n\to\infty} \|p(a_n) - p(a)\| = 0$. Let $p_k(t)$ be a sequence of polynomials such that $\lim_{k\to\infty} p_k(t) = \Phi(t)$ uniformly on $[\alpha, \beta]$. Since for any $n \in \mathbb{N}$,

$$\|\Phi(a_n) - p_k(a_n)\| = \sup_{t\in\sigma(a_n)} |\Phi(t) - p_k(t)| \le \sup_{t\in[\alpha,\beta]} |\Phi(t) - p_k(t)|,$$

we have that $\lim_{k\to\infty} \|\Phi(a_n) - p_k(a_n)\| = 0$ uniformly on $\mathbb{N}$. In the same way, we get $\lim_{k\to\infty} \|p_k(a) - \Phi(a)\| = 0$. Hence for arbitrary $\epsilon > 0$, there exists a $K \in \mathbb{N}$ such that, for any $k > K$ and any $n \in \mathbb{N}$, we have

$$\|\Phi(a_n) - p_k(a_n)\| < \frac{\epsilon}{3} \quad \text{and} \quad \|p_k(a) - \Phi(a)\| < \frac{\epsilon}{3}.$$

Taking $k_0 = K + 1$, we have $\lim_{n\to\infty} \|p_{k_0}(a_n) - p_{k_0}(a)\| = 0$ since $p_{k_0}(t)$ is a polynomial. So there is an $N \in \mathbb{N}$ such that, for any $n > N$, we have

$$\|p_{k_0}(a_n) - p_{k_0}(a)\| < \frac{\epsilon}{3}.$$

Now for any $n > N$, we obtain

$$\|\Phi(a_n) - \Phi(a)\| \le \|\Phi(a_n) - p_{k_0}(a_n)\| + \|p_{k_0}(a_n) - p_{k_0}(a)\| + \|p_{k_0}(a) - \Phi(a)\| < \epsilon.$$

Therefore, we have $\lim_{n\to\infty} \|\Phi(a_n) - \Phi(a)\| = 0$. □

Lemma 3.4 *Suppose that* (X, Σ, μ) *is a finite measure space, and suppose that* $\mathfrak{A}$ *is a unital* C^**-algebra. Let* $\{f_k\}$ *be a sequence of Bochner integrable maps from* X *to* $\mathfrak{A}_{\rm sa}$ *such that* $\sigma(f_k(x)) \subseteq [\alpha, \beta]$ *for all* $x \in X$ *and* $k \in \mathbb{N}$. *If the map* $f : X \mapsto \mathfrak{A}_{\rm sa}$ *satisfies that* $\lim_k \|f_k(x) - f(x)\| = 0$ μ*-almost everywhere, then* f *is Bochner integrable,* $\sigma(f(x)) \subseteq [\alpha, \beta]$ μ*-almost everywhere and*

$$\lim_{k\to\infty} \left\| \int_X f_k d\mu - \int_X f d\mu \right\| = 0.$$

Proof For every $\varphi \in S(\mathfrak{A})$ and $k \in \mathbb{N}$, we denote

$$F_k(x) = \varphi(f_k(x)), \quad x \in X.$$

Then F_k is measurable and bounded above by β and below by α. Because

$$\varphi(f(x)) = \lim_k \varphi(f_k(x))$$

μ-almost everywhere, we have $\sigma(f(x)) \subseteq [\alpha, \beta]$ μ-almost everywhere.

Since $\|f_k(x)\| \leq \max\{|\alpha|, |\beta|\}$ μ-almost everywhere and μ is a finite measure on X, by [7, Chapter II, Theorem 3], we have that f is Bochner integrable and

$$\lim_{k\to\infty} \left\| \int_X f_k d\mu - \int_X f d\mu \right\| = 0. \qquad \square$$

Lemma 3.5 *Let $\mathfrak{A}$ be a unital C^*-algebra with a Lip-norm L, and let (X, Σ, μ) be a measure space. If $f : X \mapsto \mathfrak{A}_{\mathrm{sa}}$ is μ-measurable and if $\sigma(f(x)) \subset (\alpha, \beta)$ for $x \in X$, then there is a sequences $\{f_k\}$ of simple maps from X to $\mathfrak{A}_{\mathrm{sa}}$ such that*

(1) *$\sigma(f_k(x)) \subseteq [\alpha, \beta]$ for all $x \in X$ and $k \in \mathbb{N}$;*

(2) *$\lim_{k\to\infty} \|f_k(x) - f(x)\| = 0$ μ-almost everywhere.*

Proof Since f is μ-measurable, there is a sequence $\{F_k\}$ of simple maps from X to $\mathfrak{A}_{\mathrm{sa}}$ such that

$$\lim_{k\to\infty} \|F_k(x) - f(x)\| = 0$$

μ-almost everywhere. We now adjust each F_k so that its spectra over all $x \in X$ contained in the closed interval $[\alpha, \beta]$. For each $k \in \mathbb{N}$, let $E_k = F_k^{-1}([\alpha 1_{\mathfrak{A}}, \beta 1_{\mathfrak{A}}])$. By Proposition 3.1, we have $E_k \in \Sigma$ for all $k \in \mathbb{N}$. Now we define the maps $f_k : X \to \mathfrak{A}_{\mathrm{sa}}$ by

$$f_k(x) = F_k(x)\chi_{E_k}(x) + \chi_{E_k^c}(x)\left(\frac{\alpha+\beta}{2}\right)1_{\mathfrak{A}}, \quad x \in X.$$

Since $F_k(x)$ is simple, it is not hard to see that $f_k(x)$ is simple and the spectrum of $f_k(x)$ is contained in $[\alpha, \beta]$ for each $x \in X$. Since $\lim_{k\to\infty} F_k(x) = f(x)$ μ-almost everywhere, by the upper semicontinuity of the spectrum, there is a subset $Z \subset X$ with $\mu(Z) = 0$, an open set $\Omega_x \subset (\alpha, \beta)$, and a positive integer N_x such that for each $x \in X \backslash Z$, we have $\sigma(F_k(x)) \subset \Omega_x$ for every $k \geq N_x$. Hence if $k \geq N_x$, then $f_k(x) = F_k(x)$ for all $x \in X \backslash Z$. Therefore, $\lim_{k\to\infty} \|f_k(x) - f(x)\| = 0$ μ-almost everywhere. $\square$

Theorem 3.6 *Let $\mathfrak{A}$ be a unital C^*-algebra with a Lip-norm L, and let (X, Σ, μ) be a probability space. Suppose that $I \subset \mathbb{R}$ is an open interval that contains $[\alpha, \beta]$. If $f : X \mapsto \mathfrak{A}_{\mathrm{sa}}$ is a Bochner integrable function for which $\sigma(f(x)) \subseteq [\alpha, \beta]$ for every $x \in X$, then*

$$\Phi\left(\int_X f d\mu\right) \leq \int_X (\Phi \circ f) d\mu$$

for every $\mathfrak{A}$-convex function Φ of order one from I to $\mathbb{R}$.

Proof From the hypothesis $\sigma(f(x)) \subseteq [\alpha, \beta]$ for every $x \in X$, we see that $\alpha \leq \varphi(f(x)) \leq \beta$ for each $\varphi \in S(\mathfrak{A})$. So for all $\varphi \in S(\mathfrak{A})$, we have

$$\alpha = \int_X \alpha d\mu(x) \leq \int_X \varphi(f(x)) d\mu(x) \leq \int_X \beta d\mu(x) = \beta.$$

By (2) in Lemma 3.2, we obtain

$$\alpha \leq \varphi\left(\int_X f(x)d\mu(x)\right) \leq \beta, \quad \varphi \in S(\mathfrak{A}).$$

This implies that $\sigma\left(\int_X f(x)d\mu(x)\right) \subseteq [\alpha, \beta]$.

Because I is an open interval that contains the closed interval $[\alpha, \beta]$, there are $\alpha' < \alpha$ and $\beta' > \beta$ such that $[\alpha, \beta] \subset [\alpha', \beta'] \subset I$. By Lemma 3.5, there is a sequence $\{f_k\}$ of simple measurable maps from X to $\mathfrak{A}_{\mathrm{sa}}$ such that $\sigma(f_k(x)) \subseteq [\alpha', \beta']$ for every $x \in X$ and $\lim_{k\to\infty} \|f_k(x) - f(x)\| = 0$ μ-almost everywhere. By Lemma 3.3,

$$\lim_{k\to\infty} \|(\Phi \circ f_k)(x) - (\Phi \circ f)(x)\| = 0$$

μ-almost everywhere. Suppose

$$f_k(x) = \sum_{i=1}^{m_k} G_{k,i}\chi_{E_{k,i}}(x), \quad x \in X, k \in \mathbb{N},$$

where $G_{k,i} \in \mathfrak{A}_{\mathrm{sa}}$ with spectrum $\sigma(G_{k,i}) \subseteq [\alpha', \beta']$ for all $k \in \mathbb{N}$ and $i = 1, 2, \ldots, m_k$, $\bigcup_{i=1}^{m_k} E_{k,i} = X$ and $E_{k,i} \cap E_{k,j} = \emptyset$ for $i \neq j$. Then

$$(\Phi \circ f_k)(x) = \sum_{i=1}^{m_k} \Phi(G_{k,i})\chi_{E_{k,i}}(x), \quad x \in X, k \in \mathbb{N}.$$

So each $\Phi \circ f_k$ is simple. Let K be a positive integer such that $\|(\Phi \circ f_K)(x) - (\Phi \circ f)(x)\| \leq 1$ μ-almost everywhere. Then we have

$$\|(\Phi \circ f)(x)\| \leq \|(\Phi \circ f_K)(x)\| + 1 \leq \sup_{x\in[\alpha',\beta']} \{|\Phi(x)|\} + 1 < +\infty$$

μ-almost everywhere. So

$$\int_X \|(\Phi \circ f)(x)\|d\mu(x) \leq \sup_{x\in[\alpha',\beta']} \{|\Phi(x)|\} + 1 < +\infty.$$

Hence $\Phi \circ f$ is Bochner integrable, and

$$\int_X (\Phi \circ f)d\mu = \lim_{k\to\infty} \int_X (\Phi \circ f_k)d\mu.$$

Denote $\alpha_{k,i} = \mu(E_{k,i})$ for $i = 1, 2, \ldots, m_k$. Because f_k is simple and $\mu(X) = 1$, we have that $\sum_{i=1}^{m_k} \alpha_{k,i} = 1$ and $\int_X f_k d\mu = \sum_{i=1}^{m_k} \alpha_{k,i}G_{k,i}$. By the condition that $\sigma(G_{k,i}) \subseteq [\alpha', \beta']$ for all i and k, we obtain

$$\sigma\left(\int_X f_k d\mu\right) \subseteq [\alpha', \beta'] \subset I.$$

Since Φ is $\mathfrak{A}$-convex of order one, we have

$$\Phi\left(\sum_{i=1}^{m_k} \alpha_{k,i}G_{k,i}\right) \leq \sum_{i=1}^{m_k} \alpha_{k,i}\Phi(G_{k,i}),$$

that is,

$$\Phi\left(\int_X f_k d\mu\right) \leq \int_X (\Phi \circ f_k)d\mu, \quad k \in \mathbb{N}.$$

By Lemma 3.4, we have $\lim_{k\to\infty}\int_X f_k d\mu = \int_X f d\mu$. So by Lemma 3.3,

$$\lim_{k\to\infty}\Phi\left(\int_X f_k d\mu\right) = \Phi\left(\int_X f d\mu\right).$$

Combining the discussion above, we get

$$\Phi\left(\int_X f d\mu\right) \le \int_X (\Phi\circ f) d\mu. \qquad \square$$

4 Continuous Operator Jensen's Inequality for Unital $\mathfrak{A}$-fields

Definition 4.1 *Let (X,Σ,μ) be a measure space, and let $\mathfrak{A}$ be a unital C^*-algebra. We say that a map $g : X \mapsto \mathfrak{A}$ is a unital $\mathfrak{A}$-field over (X,Σ,μ) if g is μ-measurable, $g^*(x)g(x)$ is Bochner integrable and $\int_X g^*(x)g(x)d\mu(x) = 1_{\mathfrak{A}}$.*

An $\mathfrak{A}$-*valued probability measure* is a map $\omega : \Sigma \mapsto \mathfrak{A}_{\rm sa}$ such that $\omega(\emptyset) = 0$, $\omega(E)$ is positive for all $E \in \Sigma$, and ω is countable additive and $\omega(X) = 1_{\mathfrak{A}}$ (see [7]). Suppose that we have a unital $\mathfrak{A}$-field g over (X,Σ,μ). We define

$$\omega_g(E) = \int_E g^*(x)g(x)d\mu(x), \quad E \in \Sigma.$$

Then ω_g is an $\mathfrak{A}$-valued probability measure on the σ-algebra (X,Σ). We will call ω_g the *vector measure induced by g*.

Definition 4.2 *Let (X,Σ,μ) be a measure space, and let $\mathfrak{A}$ be a unital C^*-algebra. Assume that $g : X \mapsto \mathfrak{A}$ is a unital $\mathfrak{A}$-field over (X,Σ,μ). Let ω_g be the vector measure induced by g. If $f : X \mapsto \mathfrak{A}_{\rm sa}$ is Bochner integrable, we define*

$$\int_X f(x)d\omega_g(x) = \int_X g^*(x)f(x)g(x)d\mu(x).$$

From the next theorem, we can see that $g^*(x)f(x)g(x)$ is Bochner integrable. So the definition above is well-defined.

Theorem 4.3 *Let (X,Σ,μ) be a measure space, and $\mathfrak{A}$ be a unital C^*-algebra with a Lip-norm L. Suppose that the map $g : X \mapsto \mathfrak{A}$ is a unital $\mathfrak{A}$-field over (X,Σ,μ). If I is an open interval such that $[\alpha,\beta] \subset I$ and if $f : X \mapsto \mathfrak{A}_{\rm sa}$ is a Bochner integrable map for which $\sigma(f(x)) \subseteq [\alpha,\beta]$ for every $x \in X$, then $g^*(x)f(x)g(x)$ is Bochner integrable and*

$$\Phi\left(\int_X f(x)d\omega_g(x)\right) \le \int_X \Phi(f(x))d\omega_g(x)$$

for every $\mathfrak{A}$-convex function Φ from I to $\mathbb{R}$.

Proof **Case 1** $0 \in [\alpha,\beta]$ and $\Phi(0) \le 0$.

Since $\int_X g^*(x)g(x)d\mu(x) = 1_{\mathfrak{A}}$ and $\sigma(f(x)) \subseteq [\alpha,\beta]$ for every $x \in X$, we have

$$\alpha 1_{\mathfrak{A}} = \int_X \alpha g^*(x)g(x)d\mu(x) \le \int_X g^*(x)f(x)g(x)d\mu(x) \le \int_X \beta g^*(x)g(x)d\mu(x) = \beta 1_{\mathfrak{A}}.$$

So $\sigma\left(\int_X f(x)d\omega_g(x)\right) \subseteq [\alpha,\beta]$.

Since $g(x)$ is μ-measurable, there exists a sequence $\{g_n\}$ of simple functions from X to $\mathfrak{A}_{\rm sa}$ such that $\lim_{n\to\infty} g_n(x) = g(x)$ μ-almost everywhere. Hence $\lim_{n\to\infty} g_n^*(x)g_n(x) = g^*(x)g(x)$

μ-almost everywhere. Assume

$$g_n(x) = \sum_{i=1}^{m_n} H_{n,i}\chi_{E_{n,i}}(x), \quad x \in X,$$

where $H_{n,i} \in \mathfrak{A}_{\mathrm{sa}}$ for all $n \in \mathbb{N}$ and $i \in \{1, 2, \ldots, m_n\}$, $\bigcup_{i=1}^{m_n} E_{n,i} = X$ with $E_{n,i} \cap E_{n,j} = \emptyset$ for $i \neq j$. So

$$\lim_{n\to\infty} \sum_{i=1}^{m_n} H_{n,i}^* H_{n,i}\chi_{E_{n,i}}(x) = g^*(x)g(x)$$

μ-almost everywhere, and hence

$$\lim_{n\to\infty} \int_X g_n^*(x)g_n(x)d\mu(x) = \int_X g^*(x)g(x)d\mu(x) = 1_{\mathfrak{A}}.$$

We may assume that

$$\int_X g_n^*(x)g_n(x)d\mu(x) = \sum_{i=1}^{m_n} \mu(E_{n,i})H_{n,i}^* H_{n,i} \leq 1_{\mathfrak{A}},$$

otherwise $\int_X g_n^*(x)g_n(x)d\mu(x)$ can be replaced by $\left\|\int_X g_n^*(x)g_n(x)d\mu(x)\right\|^{-1} \int_X g_n^*(x)g_n(x)d\mu(x)$.

Since $g^*(x)g(x)$ is Bochner integrable and $\sigma(f(x)) \subseteq [\alpha, \beta]$ for all $x \in X$, we have

$$\int_X \|g^*(x)f(x)g(x)\|\, d\mu(x) \leq \gamma \int_X \|g^*(x)g(x)\| d\mu(x) < +\infty,$$

where $\gamma = \max\{|\alpha|, |\beta|\}$. So $g^*(x)f(x)g(x)$ is Bochner integrable. Since Φ is continuous, we have $\sigma(\Phi(f(x))) = \Phi(\sigma(f(x))) \subseteq \Phi([a, b])$, i.e., $\sigma(\Phi(f(x)))$ is bounded. In the same way, we have that $g^*(x)\Phi(f(x))g(x)$ is also Bochner integrable. Therefore, using the condition that $\lim_{n\to\infty} g_n(x) = g(x)$ μ-almost everwhere and Lemma 3.4, we obtain

$$\lim_{n\to\infty} \int_X g_n^*(x)f(x)g_n(x)d\mu(x) = \int_X g^*(x)f(x)g(x)d\mu(x)$$

and

$$\lim_{n\to\infty} \int_X g_n^*(x)\Phi\left(f(x)\right) g_n(x)d\mu(x) = \int_X g^*(x)\Phi\left(f(x)\right) g(x)d\mu(x).$$

From (3) in Lemma 3.2, we have

$$\mu(E_{n,i})H_{n,i}^*\left(\int_{E_{n,i}} \frac{1}{\mu(E_{n,i})}\Phi\left(f(x)\right) d\mu(x)\right)H_{n,i} = \int_{E_{n,i}} H_{n,i}^*\Phi\left(f(x)\right) H_{n,i}d\mu(x)$$

for each $n \in \mathbb{N}$ and $i \in \{1, 2, \ldots, m_n\}$. Since

$$0 \leq \int_X g_n(x)^* g_n(x)d\mu(x) = \sum_{i=1}^{m_n} \mu(E_{n,i})H_{n,i}^* H_{n,i} \leq 1_{\mathfrak{A}},$$

we have

$$\begin{aligned}
&\Phi\left(\sum_{i=1}^{m_n} \mu(E_{n,i})H_{n,i}^* \int_{E_{n,i}} \frac{1}{\mu(E_{n,i})} f(x)d\mu(x)H_{n,i}\right) \\
&\quad \leq \sum_{i=1}^{m_n} \mu(E_{n,i})H_{n,i}^*\Phi\left(\int_{E_{n,i}} f(x)\frac{1}{\mu(E_{n,i})}d\mu(x)\right)H_{n,i}
\end{aligned}$$

by Corollary 2.4. Because $\int_{E_{n,i}} \frac{1}{\mu(E_{n,i})} d\mu = 1$, for each $n \in \mathbb{N}$ and $i = 1, \ldots, m_n$, we have

$$\Phi\left(\int_{E_{n,i}} f(x) \frac{1}{\mu(E_{n,i})} d\mu(x)\right) \leq \int_{E_{n,i}} \Phi(f(x)) \frac{1}{\mu(E_{n,i})} d\mu(x)$$

by Theorem 3.6. Using (1) and (3) in Lemma 3.2, we get

$$\begin{aligned}\sum_{i=1}^{m_n} \mu(E_{n,i}) H_{n,i}^* \left(\int_{E_{n,i}} \Phi(f(x)) \frac{1}{\mu(E_{n,i})} d\mu(x)\right) H_{n,i} &= \int_{E_{n,i}} \sum_{i=1}^{m_n} H_{n,i}^* \Phi(f(x)) H_{n,i} d\mu(x) \\ &= \int_X g_n^*(x) \Phi\left(f(x)\right) g_n(x) d\mu(x).\end{aligned}$$

Now we obtain

$$\begin{aligned}\Phi\left(\int_X g_n^*(x) f(x) g_n(x) d\mu(x)\right) &= \Phi\left(\sum_{i=1}^{m_n} \int_{E_{n,i}} H_{n,i}^* f(x) H_{n,i} d\mu(x)\right) \\ &= \Phi\left(\sum_{i=1}^{m_n} \mu(E_{n,i}) H_{n,i}^* \int_{E_{n,i}} \frac{1}{\mu(E_{n,i})} f(x) d\mu(x) H_{n,i}\right) \\ &\leq \sum_{i=1}^{m_n} \mu(E_{n,i}) H_{n,i}^* \Phi\left(\int_{E_{n,i}} f(x) \frac{1}{\mu(E_{n,i})} d\mu(x)\right) H_{n,i} \\ &\leq \sum_{i=1}^{m_n} \mu(E_{n,i}) H_{n,i}^* \left(\int_{E_{n,i}} \Phi(f(x)) \frac{1}{\mu(E_{n,i})} d\mu(x)\right) H_{n,i} \\ &= \int_X g_n^*(x) \Phi\left(f(x)\right) g_n(x) d\mu(x).\end{aligned}$$

Therefore, by Lemma 3.3, we have

$$\begin{aligned}\Phi\left(\int_X f(x) d\omega_g(x)\right) &= \Phi\left(\int_X g^*(x) f(x) g(x) d\mu(x)\right) \\ &= \lim_{n\to\infty} \Phi\left(\int_X g_n^*(x) f(x) g_n(x) d\mu(x)\right) \\ &\leq \lim_{n\to\infty} \int_X g_n^*(x) \Phi\left(f(x)\right) g_n(x) d\mu(x) \\ &= \int_X g^*(x) \Phi\left(f(x)\right) g(x) d\mu(x) \\ &= \int_X \Phi(f(x)) d\omega_g(x).\end{aligned}$$

Case 2 The general case.

Choose $c \in [\alpha, \beta]$. Denote $[\tilde{\alpha}, \tilde{\beta}] = [\alpha - c, \beta - c]$ and $\tilde{I} = \{t - c : t \in I\}$. Define

$$\tilde{\Phi}(t) = \Phi(t + c) - \Phi(c), \quad t \in \tilde{I}$$

and

$$\tilde{f}(x) = f(x) - c 1_{\mathfrak{A}}, \quad x \in X.$$

Then we have

$$0 \in [\tilde{\alpha}, \tilde{\beta}], \quad \tilde{\Phi}(0) = 0, \quad \sigma(\tilde{f}(x)) \subseteq [\tilde{\alpha}, \tilde{\beta}], \quad x \in X.$$

So by the discussion in Case 1, we have

$$\begin{aligned}
&\Phi\left(\int_X g^*(x)\tilde{f}(x)g(x)d\mu(x)+c1_{\mathfrak{A}}\right)-\Phi(c)1_{\mathfrak{A}}\\
&=\tilde{\Phi}\left(\int_X g^*(x)\tilde{f}(x)g(x)d\mu(x)\right)\\
&\leq\int_X g^*(x)\tilde{\Phi}(\tilde{f}(x))g(x)d\mu(x)\\
&=\int_X g^*(x)\Phi(\tilde{f}(x)+c1_{\mathfrak{A}})g(x)d\mu(x)-\Phi(c)\int_X g^*(x)g(x)d\mu(x)\\
&=\int_X g^*(x)\Phi(\tilde{f}(x)+c1_{\mathfrak{A}})g(x)d\mu(x)-\Phi(c)1_{\mathfrak{A}}.
\end{aligned}$$

Hence

$$\Phi\left(\int_X g^*(x)\tilde{f}(x)g(x)d\mu(x)+c1_{\mathfrak{A}}\right)\leq\int_X g^*(x)\Phi(\tilde{f}(x)+c1_{\mathfrak{A}})g(x)d\mu(x).$$

Since $\int_X g^*(x)g(x)d\mu(x)=1_{\mathfrak{A}}$ and $f(x)=\tilde{f}(x)+c1_{\mathfrak{A}}$, we have

$$\begin{aligned}
\Phi\left(\int_X g^*(x)f(x)g(x)d\mu(x)\right)&=\Phi\left(\int_X g^*(x)(\tilde{f}(x)+c1_{\mathfrak{A}})g(x)d\mu(x)\right)\\
&=\Phi\left(\int_X g^*(x)\tilde{f}(x)g(x)d\mu(x)+c1_{\mathfrak{A}}\right)\\
&\leq\int_X g^*(x)\Phi(\tilde{f}(x)+c1_{\mathfrak{A}})g(x)d\mu(x)\\
&=\int_X g^*(x)\Phi\left(f(x)\right)g(x)d\mu(x).
\end{aligned}$$

Therefore, we obtain $\Phi\left(\int_X f(x)d\omega_g(x)\right)\leq\int_X\Phi(f(x))d\omega_g(x)$, and the proof is completed. □

When Φ is $\mathfrak{A}$-convex, it is also an $M_k(\mathfrak{A})$-convex function for any $k\in\mathbb{N}$. So we have

Corollary 4.4 *Let (X,Σ,μ) be a measure space, and $\mathfrak{A}$ be a unital C^*-algebra with a Lip-norm L. Suppose that the map $g:X\mapsto M_k(\mathfrak{A})$ is a unital $M_k(\mathfrak{A})$-field over (X,Σ,μ). If I is an open interval such that $[\alpha,\beta]\subset I$ and if $f:X\mapsto M_k(\mathfrak{A})_{\mathrm{sa}}$ is a Bochner integrable map for which $\sigma(f(x))\subseteq[\alpha,\beta]$ for every $x\in X$, then $g^*(x)f(x)g(x)$ is Bochner integrable and*

$$\Phi\left(\int_X f(x)d\omega_g(x)\right)\leq\int_X\Phi(f(x))d\omega_g(x)$$

for every $\mathfrak{A}$-convex function Φ from I to $\mathbb{R}$.

Let $\mathfrak{A}$ be a unital C^*-algebra. If v is an $\mathfrak{A}$-valued probability measure on a σ-algebra (X,Σ), then for any faithful state φ on $\mathfrak{A}$, the function $\mu:\Sigma\mapsto\mathbb{R}$ defined by

$$\mu(E)=\varphi(v(E))$$

is a probability measure. Moreover, v is μ-continuous, that is, $v(E)=0$ whenever $\mu(E)=0$ for $E\in\Sigma$. In general, v is not the indefinite integral with respect to μ of an $\mathfrak{A}$-valued Bochner integrable map on X [18]. However, for $\mathfrak{A}=M_k(\mathbb{C})$, the general Radon–Nikodym theorem holds for the Bochner integral [18]. So there is a measurable map $g:X\mapsto M_k(\mathbb{C})_{\mathrm{sa}}$ such that

$$v(E)=\int_E gd\mu,\quad E\in\Sigma.$$

In particular, taking $\varphi(a) = \frac{1}{k}\mathrm{tr}(a)$ for $a \in M_k(\mathbb{C})$, we obtain a positive and measurable map $g : X \mapsto M_k(\mathbb{C})_{\mathrm{sa}}$ such that

$$v(E) = \int_E g d\mu, \quad E \in \Sigma$$

(see [8]). In this way, we get a unital $M_k(\mathbb{C})$-field $h = \sqrt{g}$ over (X, Σ, μ) with $v = \omega_h$. We know that every separable order unit space has an abundance of Lip-norms [19]. So by Corollary 4.4, we obtain the following result in [8].

Corollary 4.5 *Let (X, Σ, v) be a matrix-valued probability measure space. If I is an open interval such that $[\alpha, \beta] \subset I$ and if f is a measurable map from X to $M_k(\mathbb{C})_{\mathrm{sa}}$ for which the spectra $\sigma(f(x)) \subseteq [\alpha, \beta]$ for every $x \in X$, then*

$$\Phi\left(\int_X f dv\right) \leq \int_X (\Phi \circ f)\, dv$$

for every operator convex function $\Phi : I \mapsto \mathbb{R}$.

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